1. Simple GUI Calculator

```
import tkinter as tk

def calculate(op):
    a = float(entry1.get())
    b = float(entry2.get())

    if op == "+":
        result = a + b
    elif op == "-":
        result = a - b
    elif op == "*":
        result = a * b
    elif op == "/":
        result = a / b if b != 0 else "Cannot divide by zero"

    result_label.config(text="Result: " + str(result))

root = tk.Tk()
root.title("Simple Calculator")
root.geometry("350x300")

tk.Label(root, text="First Number").pack()
entry1 = tk.Entry(root)
entry1.pack()

tk.Label(root, text="Second Number").pack()
entry2 = tk.Entry(root)
entry2.pack()

tk.Button(root, text="Addition", command=lambda: calculate("+")).pack()
tk.Button(root, text="Subtraction", command=lambda: calculate("-")).pack()
tk.Button(root, text="Multiplication", command=lambda: calculate("*")).pack()
tk.Button(root, text="Division", command=lambda: calculate("/")).pack()

result_label = tk.Label(root, text="Result: ")
result_label.pack()

root.main loop ()
```

2.Button calculator

```
import tkinter as tk

def add():
    result.set(float(num1.get()) + float(num2.get()))

def subtract():
    result.set(float(num1.get()) - float(num2.get()))
```

```
def multiply():
    result.set(float(num1.get()) * float(num2.get()))

def divide():
    if float(num2.get()) != 0:
        result.set(float(num1.get()) / float(num2.get()))
    else:
        result.set("Error")

root = tk.Tk()
root.title("Button Calculator")
root.geometry("300x300")

tk.Label(root, text="Number 1").pack()
num1 = tk.Entry(root)
num1.pack()

tk.Label(root, text="Number 2").pack()
num2 = tk.Entry(root)
num2.pack()

tk.Button(root, text="+", command=add).pack()
tk.Button(root, text="-", command=subtract).pack()
tk.Button(root, text="*", command=multiply).pack()
tk.Button(root, text="/", command=divide).pack()

result = tk.StringVar()
tk.Label(root, textvariable=result).pack()

root.mainloop()
```

3. Menu-Based Calculator

```
import tkinter as tk

def calculate():
    a = float(first.get())
    b = float(second.get())
    op = operation.get()

    if op == "Addition":
        ans = a + b
    elif op == "Subtraction":
        ans = a - b
    elif op == "Multiplication":
        ans = a * b
    elif op == "Division":
        ans = a / b if b != 0 else "Error"

    output.config(text="Answer: " + str(ans))
```

```
root = tk.Tk()
root.title("Menu Calculator")
root.geometry("350x300")

tk.Label(root, text="Enter First Number").pack()
first = tk.Entry(root)
first.pack()

tk.Label(root, text="Enter Second Number").pack()
second = tk.Entry(root)
second.pack()

operation = tk.StringVar()
operation.set("Addition")

menu = tk.OptionMenu(
    root, operation,
    "Addition",
    "Subtraction",
    "Multiplication",
    "Division"
)
menu.pack()

tk.Button(root, text="Calculate", command=calculate).pack()

output = tk.Label(root, text="Answer:")
output.pack()

root.mainloop()
```

4. Four-Function Calculator

```
import tkinter as tk

def calculate():
    try:
        a = float(entry1.get())
        b = float(entry2.get())
        op = operator.get()

        if op == "+":
            ans = a + b
        elif op == "-":
            ans = a - b
        elif op == "*":
            ans = a * b
        elif op == "/":
            ans = a / b
```

```
        result.config(text="Result = " + str(ans))

    except:
        result.config(text="Invalid Input")

root = tk.Tk()
root.title("Four Function Calculator")
root.geometry("400x350")

tk.Label(root, text="GUI Calculator",
         font=("Arial", 18)).pack(pady=10)

entry1 = tk.Entry(root)
entry1.pack(pady=5)

entry2 = tk.Entry(root)
entry2.pack(pady=5)

operator = tk.StringVar()
operator.set("+")

tk.Radiobutton(root, text="+", variable=operator,
               value="+").pack()
tk.Radiobutton(root, text="-", variable=operator,
               value="-").pack()
tk.Radiobutton(root, text="*", variable=operator,
               value="*").pack()
tk.Radiobutton(root, text="/", variable=operator,
               value="/").pack()

tk.Button(root, text="Calculate",
          command=calculate).pack(pady=10)

result = tk.Label(root, text="Result = ")
result.pack()

root.mainloop()
```

5. Calculator with Clear Button

```
import tkinter as tk

def calculate(op):
    try:
        a = float(first.get())
        b = float(second.get())

        if op == "+":
            ans = a + b
```

```
        elif op == "-":
            ans = a - b
        elif op == "*":
            ans = a * b
        else:
            ans = a / b if b != 0 else "Cannot divide by zero"

        result.config(text="Result: " + str(ans))

    except ValueError:
        result.config(text="Enter valid numbers")

def clear():
    first.delete(0, tk.END)
    second.delete(0, tk.END)
    result.config(text="Result:")

root = tk.Tk()
root.title("Calculator")
root.geometry("350x350")

tk.Label(root, text="GUI Calculator",
         font=("Arial", 20)).pack(pady=10)

tk.Label(root, text="First Number").pack()
first = tk.Entry(root)
first.pack()

tk.Label(root, text="Second Number").pack()
second = tk.Entry(root)
second.pack()

frame = tk.Frame(root)
frame.pack(pady=15)

tk.Button(frame, text="+",
          command=lambda: calculate("+")).grid(row=0, column=0)
tk.Button(frame, text="-",
          command=lambda: calculate("-")).grid(row=0, column=1)
tk.Button(frame, text="*",
          command=lambda: calculate("*")).grid(row=0, column=2)
tk.Button(frame, text="/",
          command=lambda: calculate("/")).grid(row=0, column=3)

tk.Button(root, text="Clear", command=clear).pack(pady=10)

result = tk.Label(root, text="Result:")
result.pack()
```